



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

PL: Efektywne pod względem danych uczenie polityk agentów ucieleśnionych w cyfrowych bliźniakach otrzymanych z neuronowej rekonstrukcji scen rzeczywistych, przy ograniczonym budżecie danych rzeczywistych, na przykładzie nawigacji wizyjnej i manipulacji robotycznej

EN: Data-efficient learning of embodied policies in digital twins built by neural reconstruction of real scenes under a limited real-data budget, with visual navigation and robotic manipulation as applications

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań częściowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:



- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,
- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	<i>(Oct 2025 – Feb 2026)</i> Research target: define the problem and map the state of the art. T1.1 Systematic literature review in four areas: (a) sim-to-real transfer and domain randomization, (b) learning under distribution shift and domain adaptation, (c) neural scene reconstruction (NeRF, 3D Gaussian Splatting), (d) learning-based visual navigation. <i>D:</i> annotated bibliography and a written research-gap statement (basis of §6). T1.2 Identification of the open problem: how much real data sim-to-real transfer needs, and whether reconstructed scenes can replace real data. <i>D:</i> problem statement and preliminary research questions, discussed with the supervisor. T1.3 Doctoral School courses of semester 1. <i>D:</i> completed courses. Place: K46.



Semestr 2	(Mar – Sep 2026) Research target: formulate the research plan and select the experimental basis. T2.1 Formulation of the thesis, research questions RQ1–RQ4 and hypotheses H1–H4 with measurable decision rules and evaluation tiers A–C. <i>D:</i> §7 and §9 of this IPB. T2.2 Comparative survey of public benchmarks and datasets (e.g. ScanNet++, HM3D), simulators (e.g. Habitat-Sim) and 3D Gaussian Splatting implementations against the needs of RQ1–RQ4 (real video + reference scan, licence, compute cost). <i>D:</i> tool-selection note with a shortlist for Stage I. T2.3 Individual Research Plan agreed with the supervisor. <i>D:</i> this IPB, signed and submitted by 30.09.2026. Place: K46.
Semestr 3	(Oct 2026 – Feb 2027) Stage I – digital-twin pipeline and learning from reconstructions (RQ1). T3.1 Final choice of benchmark scenes, simulator and reconstruction method from the T2.2 shortlist; task-agnostic pipeline from a short real capture to a simulation-ready twin (camera poses, 3D Gaussian Splatting, collision mesh) with a written capture protocol; fidelity–cost report on ≥ 10 public scenes (PSNR/SSIM/LPIPS and depth error vs. capture minutes and GPU-hours); pre-registration of the evaluation protocol (§7). <i>D:</i> pipeline code, capture protocol, fidelity–cost report and evaluation protocol in the repository. Place: K46, WCSS. T3.2 Validation stand: written agreement on robot access (planned cooperation with the K29 Denali laboratory, or another PWr platform) and sensor decision (SzD Minigrant application if a sensor must be bought). <i>D:</i> agreement, by Nov 2026. T3.3 First policies trained in reconstructed vs. generic scenes (tier A). <i>D:</i> first results; decision by Jan 2027 whether they suffice for the early option, ICML 2027 (paper submitted) or not (work continues to NeurIPS 2027).
Semestr 4	(Mar – Sep 2027) Stage II – real-data budget (RQ1–RQ2, H1, H4). T4.1 Training in reconstructions vs. generic and real-data-only baselines across budgets on tiers A–B; real-robot pilot in one PWr environment (tier C). <i>D:</i> budget–performance curves, pilot results (H1 completed on tier A). Place: K46, WCSS, Denali. Without robot access by Mar 2027, the pilot moves to sem. 5; H1 is decided on tier A either way. T4.2 Prepare and submit an article to NeurIPS 2027 (Stages I–II). <i>D:</i> submission, May 2027 (§11). T4.3 Grants, international activity and mid-term: NCN PRELUDIUM application; poster at an international summer school; foreign co-author sought for P2; NAWA Bekker application for the sem. 6 visit; autoreferat (≤ 5 pages) and 15-min presentation ($\sim$mid-Oct 2027). <i>D:</i> applications, poster, autoreferat.



Semestr 5	(Oct 2027 – Feb 2028) Mid-term evaluation (Nov 2027). Stage III – predictivity and correction (RQ4, H3, H4). T5.1 SRCC of reconstructed vs. generic simulation over ≥ 10 policy variants on tier A, validated with paired rollouts on the robot (tier C). <i>D</i>: validation report (H3 completed). T5.2 Correction of appearance and dynamics from a few real rollouts; budget comparison with real-data-only learning. <i>D</i>: analysis of results. Place: K46, WCSS, Denali. T5.3 Article P2 (Stage III) submitted to CVPR 2028 (~Nov 2027; ICLR 2028, ~late Sep 2027, if ready earlier). If P1 was rejected, its revision goes to CVPR 2028 or ICML 2028 (~Jan 2028). <i>D</i>: submission(s).
Semestr 6	(Mar – Sep 2028) Stage IV – cross-scene and cross-task generalization (RQ3, H2). T6.1 (a) <i>Navigation</i>: augmentation of reconstructed scenes and representation alignment between reconstructed and real observations; evaluation on held-out scenes (tier A), real-world datasets (tier B) and uncaptured PWR or host-lab environments (tier C). (b) <i>Manipulation</i>: twins of public tabletop scenes and objects built with the unchanged Stage I pipeline; pick-and-place policies trained in a public GPU simulator (ManiSkill3) and evaluated in public real-to-sim environments (SIMPLER), with the same ablation (generic vs. twin only vs. twin + augmentation + alignment) and a real-data-only baseline; simulation-only if time is short (§12). <i>D</i>: ablation study and cross-task results (H2 completed). Place: K46, WCSS; optional validation on a manipulator of the K29 Robotics Laboratory. T6.2 Foreign research visit (3 months, NAWA Bekker; if not funded, a 5–30-day Erasmus+ short-term mobility to a European lab; hosts in §12): joint experiments. <i>D</i>: visit report. T6.3 Article P3 (Stage IV, cross-task: navigation → manipulation) submitted to ECCV 2028 (~Mar 2028) or NeurIPS 2028 (~May 2028), or RSS 2028. <i>D</i>: submission.
Semestr 7	(Oct 2028 – Feb 2029) Consolidation (RQ2, H4). T7.1 Full real-data-budget study over all baselines with real-robot validation. <i>D</i>: final results (H4 completed). T7.2 Release of code, reconstructed scenes and evaluation protocol. <i>D</i>: public repository. T7.3 Optional article P4 (consolidated study) to ICLR 2029 (~late Sep 2028) or CVPR 2029 (~Nov 2028); start of dissertation writing. <i>D</i>: submission (optional), dissertation outline.
Semestr 8	(Mar – Sep 2029) T8.1 Editing of the doctoral dissertation; final version. <i>D</i>: dissertation submitted by 30.09.2029 (§4). T8.2 Responses to reviews and camera-ready versions of pending articles. <i>D</i>: revised manuscripts.

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission



Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

30.09.2029

(rekomenrowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Modern machine learning is limited less by models than by **data in the target domain**. Deep reinforcement learning, imitation learning and large pretrained models learn strong behaviour, but only from very large amounts of experience: near-perfect point-goal navigation in simulation took billions of frames. For embodied agents (mobile robots, manipulators) such data can only come from the real world, where it is slow, expensive and sometimes risky to collect. The central question is therefore how to learn well from **a small, fixed budget of real data**.

Simulation is the usual answer, but policies trained there often fail on real inputs. This **sim-to-real gap** is a distribution shift between training and deployment data in appearance, geometry and dynamics. Domain randomization widens the training distribution by hand and still gives no guarantee for a particular target domain; training directly on real data does not scale.

Digital twins built by neural scene reconstruction (Neural Radiance Fields, 3D Gaussian Splatting) open a third path. A few minutes of real video are turned into a photorealistic, interactive model of the target scene, in which a policy can be trained at scale (*real-to-sim-to-real*). This moves the cost from collecting experience to collecting cheap captures. It raises open machine-learning questions that have not been studied systematically: how performance **scales with the real-data budget**, which **representations** bring reconstructed and real data close enough, how to generalize beyond the captured scenes **and to other tasks and embodiments**, how a few real observations can correct the simulation, and what a twin costs to build and how faithful it is.

Why this topic, and why in this discipline. The research object is a general, task-agnostic learning methodology, which places it in *information and communication technology*: data-efficient learning, learning from reconstructed data, representation learning under distribution shift, and methods for testing whether results from simulation can be trusted. **Visual navigation** is the primary testbed, because public simulators, benchmarks and real-world datasets allow controlled, reproducible measurement of the gap; trials on a real robot validate the conclusions. **Robotic manipulation** (e.g. tabletop pick-and-place in reconstructed scenes) is the generalization domain: it tests, mainly on public simulators and benchmarks, that the method is not specific to one task or robot. The topic fits the representation learning, computer vision and generative modelling work of the Department of Artificial Intelligence (K46). The computation-heavy parts can run on the GPU infrastructure available to PWr researchers (WCSS, PLGrid). The main result, a measured answer to "how much real data is enough", is useful whatever the outcome.

Potential application areas: service, assistive and warehouse robotics, inspection and delivery, flexible industrial manipulation, and rapid deployment of learned systems in new buildings and



workcells with minimal on-site data collection. More broadly, the methods apply wherever models are trained on reconstructed or synthetic data and deployed on real sensor data: other embodied agents, industrial digital twins, and perception for autonomous systems.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / **Outline of the current state of research in the topic of the doctoral dissertation**

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Learning-based navigation and its data demands. Photorealistic simulators such as Habitat [1, 2] and CARLA [3] made large-scale training of embodied agents possible. Wijmans et al. [4] reached near-perfect point-goal navigation only after 2.5 billion simulated frames, a volume out of reach on physical robots. *Navigation foundation models* (GNM [5], ViNT [6], NoMaD [7]) are trained on pooled real-robot datasets, and Open X-Embodiment [8] shows that robot learning scales with data. Such data is costly, and every new environment or platform typically needs more of it.

The sim-to-real gap. Policies trained in simulation degrade on real hardware because of differences in appearance, geometry and dynamics [9, 10]. Kadian et al. [11] showed that simulation can poorly predict real-world performance and introduced the Sim-vs-Real Correlation Coefficient (SRCC) to measure it. Truong et al. [12] found that lower-fidelity simulation can transfer better, so *which* aspects of reality to model is itself open.

Domain randomization and adaptation. Domain randomization trains policies on widely varied synthetic appearance [13, 14] or dynamics [15], so that reality is just another variation, as in agile flight [16] and drone racing [17]. Its drawbacks are hand-tuned ranges and lower target-domain performance. Adaptive approaches fit simulation parameters to real rollouts [18], and domain adaptation [19] reduces, but does not remove, the need for real data.

Neural scene reconstruction and real-to-sim-to-real. Neural Radiance Fields [20] and 3D Gaussian Splatting (3DGS) [21] render photorealistic novel views from ordinary images, 3DGS in real time. This allows *digital-twin* simulators built from real data; NeRF2Real [22] trained bipedal skills inside a NeRF of the target scene. Since 2024 the idea has reached navigation: 3DGS twins transferred visual drone [24–26] and RGB-only legged [27] navigation policies directly to reality, Vid2Sim [28] built urban simulators from monocular video (a 68.3% real-world success-rate gain over prior simulators), and ReaDy-Go [29] added moving humans. Closest to this dissertation, EmbodiedSplat [30] fine-tunes pretrained image-goal policies in meshes from 20–30-minute phone captures, improves real-robot success in a captured scene and reports high SRCC. These systems use a fixed capture per scene: none varies the amount of real data or compares against policies trained on real robot experience at matched budgets.

Digital twins for manipulation. RialTo [23] robustifies manipulation policies by reinforcement learning in twins built from little real data; its ablation over 0–15 real demonstrations is the closest analysis of a real-data budget. SplatSim [36] renders simulation with Gaussian splats and transfers RGB manipulation policies zero-shot (86.25% real success vs. 97.5% for policies trained on real data). SIMPLER [37] shows that simulated evaluation of manipulation policies correlates strongly with real evaluation, and the GPU simulator ManiSkill3 [38] includes real-world digital-twin environments. Each work builds its twin for one task and embodiment; whether one pipeline serves navigation and manipulation, and at what capture and compute cost, has not been measured.

Learning under distribution shift. In machine-learning terms the sim-to-real gap is a domain shift. Unsupervised domain adaptation aligns feature distributions adversarially [31] or at pixel and feature

level [32]. Scaling laws [33] describe how performance grows with data, but not how *real* and *reconstructed* data trade off. 3D scene datasets (HM3D [34], ScanNet++ [35]) allow a controlled study, with real captures as a proxy for reality.

Research gap. There is no systematic study of the **trade-off between real-data budget and deployed policy performance** in real-to-sim-to-real learning, nor a twin-building protocol with measured cost and fidelity across tasks. Also missing are methods that (i) combine twins with augmentation and representation alignment to generalize to uncaptured scenes and new tasks, and (ii) correct the twin from a few real rollouts. This dissertation addresses that gap in navigation, with manipulation as the generalization domain.

References

- [1] M. Savva et al., "Habitat," ICCV, 2019.
- [2] A. Szot et al., "Habitat 2.0," NeurIPS, 2021.
- [3] A. Dosovitskiy et al., "CARLA," CoRL, 2017.
- [4] E. Wijmans et al., "DD-PPO," ICLR, 2020.
- [5] D. Shah et al., "GNM," ICRA, 2023.
- [6] D. Shah et al., "ViNT," CoRL, 2023.
- [7] A. Sridhar et al., "NoMaD," ICRA, 2024.
- [8] Open X-Embodiment Collaboration, "Open X-Embodiment," ICRA, 2024.
- [9] W. Zhao et al., "Sim-to-Real Transfer in Deep Reinforcement Learning for Robotics," IEEE SSCI, 2020.
- [10] S. Höfer et al., "Sim2Real in Robotics and Automation," IEEE T-ASE, 2021.
- [11] A. Kadian et al., "Sim2Real Predictivity," IEEE RA-L, 2020.
- [12] J. Truong et al., "Rethinking Sim2Real," CoRL, 2022.
- [13] J. Tobin et al., "Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World," IROS, 2017.
- [14] J. Tremblay et al., "Training Deep Networks with Synthetic Data," CVPRW, 2018.
- [15] X. B. Peng et al., "Sim-to-Real Transfer of Robotic Control with Dynamics Randomization," ICRA, 2018.
- [16] A. Loquercio et al., "Learning High-Speed Flight in the Wild," Sci. Robot., 2021.
- [17] E. Kaufmann et al., "Champion-level drone racing using deep reinforcement learning," Nature, 2023.
- [18] Y. Chebotar et al., "Closing the Sim-to-Real Loop," ICRA, 2019.
- [19] K. Bousmalis et al., "Using Simulation and Domain Adaptation to Improve Efficiency of Deep Robotic Grasping," ICRA, 2018.
- [20] B. Mildenhall et al., "NeRF," ECCV, 2020.
- [21] B. Kerbl et al., "3D Gaussian Splatting for Real-Time Radiance Field Rendering," ACM TOG, 2023.
- [22] A. Byravan et al., "NeRF2Real," ICRA, 2023.
- [23] M. Torne et al., "Reconciling Reality through Simulation," RSS, 2024.
- [24] A. Quach et al., "Gaussian Splatting to Real World Flight Navigation Transfer with Liquid Networks," arXiv:2406.15149, 2024.
- [25] J. Low et al., "SOUS VIDE," IEEE RA-L, 2025.
- [26] Q. Chen et al., "GRaD-Nav," IROS, 2025.
- [27] S. Zhu et al., "VR-Robo," IEEE RA-L, 2025.
- [28] Z. Xie et al., "Vid2Sim," CVPR, 2025.
- [29] S. Yoo et al., "ReaDy-Go," IEEE RA-L, 2026.
- [30] G. Chhablani et al., "EmbodiedSplat," ICCV, 2025.
- [31] Y. Ganin et al., "Domain-Adversarial Training of Neural Networks," JMLR, 2016.
- [32] J. Hoffman et al., "CyCADA," ICML, 2018.
- [33] J. Kaplan et al., "Scaling Laws for Neural Language Models," arXiv:2001.08361, 2020.
- [34] S. K. Ramakrishnan et al., "Habitat-Matterport 3D Dataset," NeurIPS Datasets and Benchmarks, 2021.
- [35] C. Yeshwanth et al., "ScanNet++," ICCV, 2023.



- [36] M. N. Qureshi et al., "SplatSim," ICRA, 2025.
[37] X. Li et al., "Evaluating Real-World Robot Manipulation Policies in Simulation," CoRL, 2024.
[38] S. Tao et al., "ManiSkill3," RSS, 2025.

7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Thesis. Embodied policies learned in a digital twin reconstructed from a small budget of real observations, with aligned representations, perform in the target domain not worse than policies learned from $\geq 10\times$ more real data, and the method transfers across scenes and tasks. Visual navigation is the primary testbed; robotic manipulation tests generalization.

Core and extensions. The dissertation stands on **RQ1–RQ2 with H1 and H4** (real-data budget, navigation). RQ3/H2 (cross-scene and cross-task generalization) and RQ4/H3 (correction, predictivity) are extensions.

RQ1. Does training an embodied policy in a twin built from minutes of real video of the target scene (*real-to-sim*), then deploying it on real observations (*sim-to-real*), close the appearance and geometry gap?

RQ2. How does target-domain performance scale with the real-data budget, and what budget suffices compared with learning from real data alone?

RQ3. Which augmentation and representation-alignment objectives let policies generalize to scenes that were *not* captured and to another task and embodiment (navigation $\rightarrow$ manipulation)?

RQ4. Can a few real rollouts correct the twin so that it predicts real performance?

Evaluation. *Navigation:* (A, primary) ≥ 10 public indoor scenes with real video and a reference 3D scan that serves as the target domain ("proxy reality"), some held out as uncaptured; (B) real-world datasets for representation metrics; (C) real-robot validation in ≥ 2 PWR environments, 20 start–goal pairs $\times$ 3 trials per policy. Metrics: success rate (SR; primary), SPL, collisions. *Manipulation:* tabletop pick-and-place in twins of public scenes and objects in a public simulator; metric: task SR. *Budget* B = minutes of target-domain data (capture plus real experience). Baselines: a pretrained policy fine-tuned by behaviour cloning on target-domain data of budget B (*real-data-only*), and the same policy trained with domain randomization on generic scenes (*generic*). Tests: one-sided ($\alpha = 0.05$), bootstrapped over scenes or start–goal pairs, Holm-corrected per hypothesis, pre-registered in the repository.

H1 (RQ1, core). A navigation policy trained only in the twin (a) beats the generic baseline in SR and (b) is **non-inferior** to the real-data-only baseline with a $\geq 10\times$ larger budget: the lower 95% bound of $SR_{\text{twin}} - SR_{\text{real}}$ is above -10 pp on tier A and above -15 pp on tier C (120 pooled episodes per policy; $\approx 90\%$ power at $SR \approx 0.8$).

H2 (RQ3). In uncaptured navigation scenes, and in manipulation with an unchanged pipeline, twin + augmentation + representation alignment reaches a higher SR than the twin alone and than the generic baseline, and alignment lowers the feature distance between reconstructed and real frames (tier B).

H3 (RQ4). Over ≥ 10 navigation policy variants, the Sim-vs-Real Correlation Coefficient (SRCC) of the twin is higher than that of a generic simulator (95% CI of the difference excludes 0) and does not drop after refinement from real rollouts (lower 95% bound of the change ≥ -0.1).

H4 (RQ2, core). The full pipeline reaches a target navigation SR τ (best real-data-only SR minus 10 pp) with $B_{\text{pipeline}}(\tau) \leq 0.1 \cdot B_{\text{real}}(\tau)$: the upper 95% bound of the ratio, from fitted budget–SR scaling



curves, is ≤ 0.1 . If real-data-only training misses τ within the largest feasible budget, the ratio is reported as a bound.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The dissertation is carried out in **information and communication technology** (*informatyka techniczna i telekomunikacja*), in the area of machine learning. Its object is a general, task-agnostic methodology for learning embodied policies in **digital twins** built by neural reconstruction of real scenes (real $\rightarrow$ twin $\rightarrow$ policy $\rightarrow$ real). Visual navigation is the primary testbed and robotic manipulation the generalization domain; control design and mechanical engineering are not the object of the research.

Key original contribution (core, RQ1–RQ2, H1 + H4). The first systematic, quantitative characterization, to our knowledge, of how performance of policies learned in **neural-reconstruction twins** scales with the **real-data budget** of the target domain. It answers "how much real data is enough" with measured budget–success-rate scaling curves and a non-inferiority comparison against learning from real data alone, on public navigation benchmarks and validated on a real robot.

Contributions to machine learning methodology:

1. **Data-efficient learning under a real-data budget (core; RQ2, H4).** A training procedure that treats the amount of real target-domain data (captures and a few real rollouts) as an explicit budget, and scaling curves that quantify the trade-off between real data and deployed performance.
2. **Learning from reconstructed data (core; RQ1, H1).** Evidence of when training in photorealistic twins (3D Gaussian Splatting for appearance, extracted geometry for collisions) closes the distribution gap to real observations, compared with generic simulation and domain randomization.
3. **Cross-scene and cross-task generalization (extension; RQ3, H2).** Augmentation of reconstructed scenes combined with representation alignment between reconstructed and real observations, so that policies generalize to scenes that were never captured, and evidence that the same pipeline transfers from navigation to manipulation. It connects the digital-twin and domain-randomization lines of research.
4. **Closed-loop correction and evaluation methodology (extension; RQ4, H3).** A method to update the twin from a few real rollouts, and a protocol for measuring how well simulation predicts real performance (SRCC, pre-specified margins and decision rules), so that sim-to-real claims are tested, not asserted.

Contributions in engineering terms (ITiT):

5. **Digital-twin pipeline, software and data (supports all RQs).** A reusable, task-agnostic pipeline and protocol that turns a short RGB(-D) capture into a simulation-ready twin, with measured cost (capture minutes, compute hours) and fidelity metrics; released code, reconstructed scenes and evaluation sets with fixed episodes for reproducible sim-to-real benchmarking in navigation and manipulation.

Significance for the discipline. Target-domain data is the main cost of deploying learned systems. Showing how far cheap captures and reconstruction can replace it, across tasks and embodiments, and which representations make this work, is relevant to machine learning under distribution shift in general. The evaluation methodology (item 4) is reusable for other simulators and embodied-AI tasks.



Dissemination. Results are planned for peer-reviewed conferences worth 200 points and assigned to ITiT on the ministerial list of 5.01.2024: NeurIPS, ICML and ICLR (machine learning), CVPR, ICCV and ECCV (computer vision), and RSS (robot learning), as in §3, §11 and §12; the cross-task (manipulation) study is paper P3.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The research follows four stages (I–IV) matching §3; for each we give the method, data, metric and success criterion. The pipeline (real → twin → policy → real) is task-agnostic: visual navigation is the primary testbed, where H1, H3 and H4 are decided on public benchmarks and datasets and validated on a real robot (the three evaluation tiers of §7); manipulation tests cross-task generalization (H2, Stage IV). Stages I–II (H1 and the real-data budget, RQ1–RQ2) are the core; Stages III–IV extend it.

Data and evaluation tiers. *Tier A (benchmark, primary):* ≥ 10 public indoor scenes that have both real video and a reference 3D scan, e.g. ScanNet++ (laser scans, DSLR images and iPhone RGB-D video of real rooms); the reference scan, loaded into a GPU simulator (Habitat or Isaac Sim / Isaac Lab, chosen in T3.1), is the target domain, and some scenes are held out as uncaptured. Large public scene sets (e.g. HM3D) train the generic baseline. *Tier B (real-world datasets):* held-out real frames and trajectories from public navigation datasets (e.g. SCAND, RECON) for representation metrics and offline checks. *Tier C (real-robot validation):* ≥ 2 PWR environments with a mobile robot (planned cooperation with the K29 "Denali" Autonomous Robots Laboratory: Pioneer 3-DX and DrRobot Jaguar 4x4 under ROS 2; agreement in T3.2), with an RGB-D camera and wheel odometry; a sensor is bought from an SzD Minigrant if needed. The **real-data budget** B (minutes of target-domain data: capture video plus real experience) is an explicit experimental variable: on tier A it is set by subsampling the scene's video and the number of rollouts in the reference scan. *Estimated robot time:* about 16 robot-hours per validation campaign (4 policies $\times$ 120 episodes at ~ 2 min per episode including reset), in sem. 4–5 and sem. 7. Compute: WCSS Lem (NVIDIA H100) and PLGrid allocations (to be applied for), plus departmental GPUs.

Stage I – digital-twin pipeline and learning from reconstructions (RQ1; sem. 3).

- *Method:* one task-agnostic pipeline from a short RGB(-D) capture to a simulation-ready twin: camera poses from Structure-from-Motion (COLMAP) or the dataset's poses; appearance from 3D Gaussian Splatting (NeRF as baseline); collision geometry from depth fusion or mesh extraction; export to the simulator with rendering from the reconstruction and physics from the mesh. The same pipeline serves rooms (navigation) and tabletop scenes and objects (manipulation).
- *Protocol and cost:* a written capture protocol (camera path, duration, lighting, coverage check); for each twin we log capture minutes, operator minutes and GPU-hours of reconstruction and mesh extraction.
- *Fidelity metrics and success criterion:* PSNR, SSIM and LPIPS on held-out views; depth error and Chamfer distance of the collision mesh to the reference scan. A scene passes if its held-out PSNR is within a margin X dB of the value reported for 3DGS [21] on comparable indoor scenes (X fixed in the repository before Stage II) and its collision mesh has no holes on the evaluation paths. Fidelity–cost curves (fidelity vs. capture minutes) are reported and later related to the transfer gap (Stage III).

Stage II – learning under a real-data budget (RQ1–RQ2, H1, H4; sem. 3–4).

- *Method:* (i) deep reinforcement learning (PPO / DD-PPO) for point-goal and image-goal navigation, trained at scale in the reconstructions; (ii) imitation learning and fine-tuning of pretrained navigation models (GNM, ViNT, NoMaD) on trajectories generated in simulation. Input: RGB(-D); output: velocity commands.



- *Baselines (§7)*: the same policy trained on generic public scenes with domain randomization; the real-data-only baseline, a pretrained model fine-tuned by behaviour cloning on target-domain trajectories of budget B, at matched and $\geq 10\times$ larger budgets; pretrained models zero-shot.
- *Metric and success criterion*: success rate (primary), SPL and collisions as a function of B; budget–SR scaling curves fitted per method; H1 and H4 decision rules in §7 on tier A, with the robot pilot (tier C, one environment) as the first validation.

Stage III – predictivity and correction (RQ4, H3, H4; sem. 4–5).

- *Validation protocol*: paired rollouts in simulation and in the target domain on the same start–goal pairs. Over ≥ 10 policy variants we compute the **Sim-vs-Real Correlation Coefficient (SRCC)** between simulated and target-domain success/SPL for the reconstructed and the generic simulator, on tier A and on the robot (tier C), and relate per-scene fidelity (Stage I) to the transfer gap.
- *Method*: the discrepancy between target-domain and replayed trajectories updates the scene (re-capture of poorly reconstructed regions) and fits dynamics and sensor parameters (system identification, adaptive randomization in the style of SimOpt); the policy is then fine-tuned in the corrected simulation.
- *Metric and success criterion*: SRCC with bootstrap confidence intervals; target-domain rollouts needed to reach the target success rate compared with real-data-only learning; H3 and H4 decision rules in §7.
- *Fallback* (low SRCC or no gain): reported as a negative finding, with the error attributed to appearance, geometry or dynamics by ablation; Stage II conclusions do not depend on this stage.

Stage IV – cross-scene and cross-task generalization (RQ3, H2, H4; sem. 6–7).

- *Method*: augmentation of reconstructed scenes (object insertion and rearrangement, lighting and appearance randomization, sensor noise, simulated pedestrians) and **representation alignment**: an auxiliary objective that brings encoder features of reconstructed and real observations of the same place together (e.g. contrastive or distribution-matching losses). Ablation: generic + randomization vs. reconstruction only vs. reconstruction + augmentation vs. + alignment.
- *Data*: held-out uncaptured scenes (tier A), real frames (tier B), and uncaptured real environments at PWr and, during the foreign visit, at the host lab (tier C).
- *Cross-task study (manipulation, sem. 6)*: the Stage I pipeline, unchanged, builds twins of public tabletop scenes and objects; pick-and-place policies are trained in them by imitation and reinforcement learning in the GPU simulator ManiSkill3 [38] and evaluated in public real-to-sim environments (SIMPLER [37], whose simulated results correlate with real robots) as proxy reality. Same ablation as for navigation (generic vs. twin only vs. twin + augmentation + alignment), plus behaviour cloning on public real demonstrations (Open X-Embodiment [8]) at matched budget. Metric: task success rate (H2 rule in §7). Real validation on a PWr manipulator is optional.
- *Metric and success criterion*: success rate and SPL in uncaptured scenes, and the feature-distribution distance between reconstructed and real frames (H2 rule in §7); the full budget curve over all baselines (H4, sem. 7).

Statistics and reproducibility. All comparisons follow the §7 protocol, which is pre-registered in the project repository before each stage; changes are logged with reasons. Code, configurations and reconstructed scenes are versioned and released where the dataset licences permit.

10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the



doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).

Streszczenie popularnonaukowe

Współczesna sztuczna inteligencja uczy się na przykładach i zwykle potrzebuje ich ogromnej liczby. W wielu zastosowaniach dobre przykłady pochodzą jednak tylko z prawdziwego świata, a ich zbieranie jest powolne, drogie, a czasem ryzykowne. Dobrym przykładem są roboty, na przykład w magazynach, szpitalach, biurach czy fabrykach. Aby robot nauczył się samodzielnie poruszać lub chwycić przedmioty, potrzebuje milionów prób i błędów. Dlatego uczy się go zwykle w symulacji komputerowej, czyli w wirtualnym świecie przypominającym grę. Świat wirtualny różni się jednak od prawdziwego: inaczej wyglądają ściany, światło i przedmioty. Model sztucznej inteligencji wyuczony w symulacji często zawodzi, gdy zobaczy prawdziwe obrazy.

Celem rozprawy jest opracowanie metod uczenia maszynowego, które pozwolą sztucznej inteligencji dobrze działać w prawdziwym świecie przy bardzo małej ilości prawdziwych danych. Pomysł polega na nagraniu krótkiego filmu z danego miejsca, na przykład korytarza, biura lub stołu roboczego. Na jego podstawie metody sztucznej inteligencji odtwarzające trójwymiarowe sceny tworzą wierną, fotorealistyczną cyfrową kopię tego miejsca, tzw. cyfrowego bliźniaka. Opracujemy uniwersalny sposób budowania takich kopii i zmierzmy, ile kosztuje ich wykonanie i jak wiernie odwzorowują rzeczywistość. W tej kopii model może bezpiecznie i szybko ćwiczyć miliony razy. Zbadamy też, jak nauczyć model takiego „widzenia” świata, by obrazy z kopii i z rzeczywistości były dla niego podobne, oraz jak kilka prawdziwych prób może poprawić symulację. Głównym zadaniem testowym będzie nawigacja robota mobilnego. Metody zostaną sprawdzone przede wszystkim na publicznie dostępnych wirtualnych budynkach i zbiorach prawdziwych nagrań, a następnie potwierdzone na prawdziwym robocie w budynkach Politechniki Wrocławskiej. Aby pokazać, że podejście działa także w innych zadaniach, zastosujemy je również do chwytania i przenoszenia przedmiotów przez ramię robota, przede wszystkim w publicznie dostępnych symulatorach.

Badania pokażą, ile prawdziwych danych naprawdę potrzeba, aby wyuczony model działał niezawodnie, oraz jak sprawić, by radził sobie również w miejscach, których wcześniej nie widział, i w innych zadaniach. Spodziewanym efektem jest znaczne obniżenie kosztu i czasu wdrażania systemów sztucznej inteligencji w nowych miejscach, co może przyspieszyć wykorzystanie robotów w logistyce, opiece i przemyśle. Opracowane oprogramowanie i dane zostaną udostępnione publicznie.

Abstract for general public

Modern artificial intelligence learns from examples, and it usually needs an enormous number of them. In many applications, however, good examples come only from the real world, where collecting them is slow, expensive and sometimes risky. Robots, for example in warehouses, hospitals, offices or factories, are a good example. For a robot to learn to move or grasp objects on its own, it needs millions of trials and errors. That is why it is usually trained in a computer simulation, a virtual world similar to a video game. The virtual world, however, differs from the real one: walls, light and objects look different. An artificial intelligence model trained in simulation often fails when it sees real images.

The goal of this dissertation is to develop machine learning methods that let artificial intelligence work well in the real world using very little real data. The idea is to record a short video of a place, such as a corridor, an office or a workbench. From this video, artificial intelligence methods that reconstruct three-dimensional scenes create a faithful, photorealistic digital copy of the place, a so-called digital twin. We will develop a universal way of building such copies and measure how much they cost to make and how faithfully they reproduce reality. In this copy the model can practice safely and quickly millions of times. We will also study how to teach the model to "see" the world so that images from the copy and from reality look similar to it, and how a few real trials can correct the simulation. Mobile-robot navigation will be the main test task. The methods will be tested mainly on publicly available virtual buildings and collections of real recordings, and then confirmed on a real robot in the buildings of Wrocław University of Science and Technology. To show that the approach



also works in other tasks, we will also apply it to grasping and moving objects with a robot arm, mainly in publicly available simulators.

The research will show how much real data is truly needed for a trained model to work reliably, and how to make it cope with places it has never seen before and with other tasks. The expected result is a significant reduction in the cost and time of deploying artificial intelligence systems in new places, which may speed up the use of robots in logistics, care and industry. The software and data developed will be made publicly available.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

May 2027 / maj 2027

Paper submitted for review to the **Conference on Neural Information Processing Systems (NeurIPS 2027)**, peer-reviewed international conference proceedings worth **200 points** on the ministerial list (conference Lp. 87), assigned to the discipline *informatyka techniczna i telekomunikacja*. It reports Stages I–II (learning from reconstructed scenes and the first real-data-budget curve on public benchmarks; task T4.2). If the results are ready earlier, the paper goes to **ICML 2027** (expected deadline late January 2027, also 200 points). If it is rejected, the revised paper is resubmitted to the next 200-point conference (§12).

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

Planned outputs by year. All papers target 200-point conferences assigned to the discipline (ministerial list of 5.01.2024; points to be re-checked against the new list expected in early 2027).

Year	Papers (planned submission, 200 points each)	Grants	Mobility, events
1 (2025/26)	—	—	—



2 (2026/27)	P1, Stages I–II: NeurIPS 2027, May 2027 (§11); early option ICML 2027	SzD Minigrant; NCN PRELUDIUM	Summer school poster
3 (2027/28)	P2 (Stage III): CVPR or ICLR 2028; P3 (Stage IV, cross-task): ECCV, NeurIPS or RSS 2028	NAWA Bekker	Mid-term; 3-month visit (sem. 6)
4 (2028/29)	Optional P4, consolidated study: ICLR 2029 or CVPR 2029	—	Code and data release; dissertation

P1–P3 are the thematically linked core of the dissertation; P4 is optional.

Foreign research visit (candidate hosts, not yet contacted). (1) Multi-robot Systems Lab, Stanford University (M. Schwager): Gaussian-splatting scene models and navigation. (2) Z. Kira's group, Georgia Tech: EmbodiedSplat [30]. A European host (which also opens the Erasmus+ 5–30-day route) is sought with the supervisor in sem. 4, with a foreign co-author for P2.

Collaboration at PWr. Real-robot validation is planned with the *Denali* Autonomous Robots Laboratory (K29, W12N; Pioneer 3-DX, Jaguar 4x4, ROS 2); a written agreement is task T3.2 (Nov 2026). Optional manipulation validation may use K29's Robotics Laboratory arms. Consultations on 3D reconstruction with K46's genwro.AI group are possible.

Risks and mitigation (semester affected).

- *Rejection at a top venue* (4–7): every paper has a resubmission path inside the 200-point set, to the next deadline 2–4 months later (e.g. NeurIPS → ICLR or CVPR → ICML → ECCV/ICCV), after revision. With three core papers and several yearly deadlines, the degree requirement does not depend on one decision.
- *Changing venue points* (sem. 3–4): the new list may change points; venues are re-checked when it appears, and the plan keeps several 200-point alternatives per paper.
- *No robot access or sensor* (3–5): hypotheses are decided on public benchmarks and datasets; the robot only validates, so it can move by a semester or to another PWr robot.
- *Poor reconstruction of low-texture or large scenes* (3–4): depth priors, scene splitting, other scenes from the public dataset.
- *Sim-to-real predictivity fails* (4–5): pre-specified SRCC protocol; a negative result still answers RQ4.
- *Scope creep from manipulation* (6–7): a generalization study on the Stage I pipeline and public simulators; if time is short, it is reduced to simulation-only experiments.

Prior work (background only). A pre-PhD NLP paper (ACL 2025 SRW) gives experience in representation learning; it is not part of the dissertation.

Ethics and data. Own captures avoid or anonymize personal data (GDPR, PWr rules); public datasets are used under their licences.

13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

- **Weekly meeting** (~1 h, in person at K46 or online via MS Teams) to discuss progress, results and next steps.



- **Monthly written progress summary** (completed tasks, results, problems, plan). It lists every IPB task from §3 with its **% completion**, in the same form as the task table of the mid-term self-report.
- **Semester review** against the IPB schedule (§3), before each annual report and the mid-term evaluation. Deviations from the plan and their reasons are recorded and agreed with the supervisor.
- **Supervisor co-signs** the annual reports and the mid-term self-report (autoreferat).
- **Paper-writing sprints** before each submission deadline, with joint editing in Overleaf. The supervisor co-authors and reviews the papers before submission.
- **Ongoing communication** via e-mail / Teams chat. The code and experiment logs are in a shared Git repository that the supervisor can access.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor